

WESTERN HEAD, NS, Canada

WMO: 714110

Lat: 43.990N

Lon: 64.664W

Elev: 33

StdP: 14.68

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.1	9.1	-5.0	4.2	8.2	-1.0	5.2	12.2	35.6	41.4	31.3	40.0	9.1	300	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.5	77.0	64.5	72.6	63.6	69.3	63.2	68.2	72.0	66.2	69.6	64.7	67.7	10.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	98.9	69.8	65.0	92.7	67.5	63.5	87.9	65.9	32.5	72.0	30.9	69.7	29.8	67.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.0	23.9	20.8	DB	-0.6	87.1	5.1	4.3	-4.3	90.2	-7.3	92.7	-10.2	95.1	-13.9	98.3	
			WB	-0.7	73.9	4.9	3.1	-4.3	76.1	-7.2	77.9	-10.0	79.6	-13.6	81.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	26.8	27.3	32.5	40.3	47.6	55.2	60.2	61.6	58.8	50.7	41.7	33.0
	DBStd	13.89	9.35	8.17	6.97	5.06	5.05	5.23	4.70	4.13	5.02	6.25	7.30	8.36
	HDD50	3167	720	635	544	292	109	9	0	0	2	69	260	527
	HDD65	7440	1185	1055	1007	740	539	300	163	121	194	445	699	992
	CDD50	1245	0	0	1	2	35	164	316	360	267	89	11	0
	CDD65	44	0	0	0	0	0	4	14	16	9	1	0	0
	CDH74	291	0	0	2	0	12	57	122	70	25	3	0	0
	CDH80	67	0	0	0	0	2	14	34	13	4	0	0	0
Wind	WSAvg	9.5	11.1	10.8	10.8	10.0	8.5	8.1	7.3	7.5	8.6	10.4	10.6	11.0
Precipitation	PrecAvg	55.3	4.9	5.2	4.8	4.3	4.4	3.3	2.9	4.3	3.9	4.7	6.3	6.3
	PrecMax	76.8	8.7	9.2	7.6	6.9	7.8	5.8	7.6	8.6	12.8	10.6	14.3	9.7
	PrecMin	34.4	1.5	2.6	1.3	1.1	1.3	1.7	0.8	1.2	0.9	1.9	2.4	3.6
	PrecStd	10.8	1.6	2.0	1.7	1.5	2.0	1.1	1.8	2.4	3.0	2.4	3.0	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.8	47.4	56.1	62.9	73.7	80.7	84.6	81.0	77.2	70.2	60.0	53.2
		MCWB	46.6	43.3	45.3	49.2	57.5	63.9	67.8	67.3	66.6	60.2	55.3	51.3
	2%	DB	45.5	43.2	47.7	55.7	64.2	73.0	76.7	75.0	72.2	64.9	56.6	50.3
		MCWB	43.6	40.9	41.0	45.5	53.4	59.7	64.7	65.4	65.2	60.1	54.0	49.0
	5%	DB	43.0	40.5	43.8	50.6	58.7	67.0	70.8	70.8	69.1	62.5	54.5	47.9
		MCWB	41.2	38.4	39.9	44.3	50.9	58.2	64.3	64.2	63.5	59.2	52.0	46.5
	10%	DB	40.3	38.3	41.2	47.4	54.5	62.3	67.0	68.4	66.6	60.5	52.7	45.3
		MCWB	38.3	36.3	38.3	43.2	49.5	57.3	63.2	64.1	62.3	57.4	50.4	43.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	44.9	47.8	52.6	61.4	66.4	72.1	71.6	69.9	64.3	57.6	52.3
		MCDB	48.0	46.5	51.9	57.1	68.3	76.7	79.8	77.7	73.4	66.6	58.2	52.7
	2%	WB	44.2	41.6	43.2	48.2	55.2	62.3	67.3	68.0	67.3	62.0	55.0	49.3
		MCDB	44.9	42.7	45.7	52.0	61.2	68.6	72.0	71.2	69.9	63.6	55.8	49.9
	5%	WB	41.5	38.9	40.9	46.0	52.2	59.9	65.2	66.1	65.4	60.1	53.0	46.9
		MCDB	42.5	40.0	42.7	49.3	56.4	64.7	68.9	69.1	67.7	61.7	53.9	47.7
	10%	WB	38.6	36.5	38.6	44.0	50.1	57.9	63.4	64.6	63.5	58.2	50.9	44.1
		MCDB	40.0	38.0	40.7	46.6	53.4	61.6	66.7	67.4	65.7	59.9	52.2	45.1
Mean Daily Temperature Range	5% DB	MDBR	14.2	13.9	13.1	13.0	12.8	13.2	13.2	13.5	14.4	14.5	13.0	13.3
		MCDBR	15.6	14.2	16.1	19.9	22.2	23.8	21.8	18.7	17.4	16.1	13.6	14.3
		MCWBR	15.7	13.0	12.2	12.9	13.3	13.1	12.8	11.5	11.5	11.9	12.4	14.2
	5% WB	MCDBR	15.5	13.7	14.5	15.8	17.8	19.2	17.4	14.9	13.6	13.2	12.6	13.7
		MCWBR	16.0	13.4	12.6	12.0	12.2	12.3	12.2	10.6	10.3	11.3	12.7	14.3
Clear-Sky Solar Irradiance	taub		0.306	0.304	0.332	0.359	0.380	0.406	0.423	0.402	0.363	0.341	0.323	0.308
	taud		2.487	2.513	2.456	2.423	2.406	2.364	2.337	2.390	2.461	2.528	2.534	2.509
	Ebn at Noon		262	285	288	287	283	275	268	271	274	266	253	249
	Edh at Noon		22	26	31	35	36	38	39	36	31	25	21	20
All-Sky Solar Radiation	RadAvg		444	724	1121	1397	1600	1763	1733	1602	1326	855	489	360
	RadStd		33	61	89	126	194	169	120	108	77	93	53	31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.67	N/A	N/A	N/A	N/A	+187	+22
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	+0.76	+0.99	N/A	N/A	+104	+13

Nomenclature: See separate page